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AI-Driven Sheet Metal Part Quality Inspection

(AI-Driven Visual Inspection System for Sheet Metal Bracket Quality Control)

Automated Defect Detection, Classification, Reporting and Root-Cause Analysis Support using Deep Learning and Machine Vision

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1. Project Overview

Sheet-metal parts can undergo cutting, punching, bending and welding before they are ready for further processing or shipment. At these stages, defects such as burrs, hole-position deviations, dents and bend-angle deviations can occur. Many small and mid-sized fabrication shops rely on manual visual inspection and on manual measurement with callipers or gauges for quality checking. Manual inspection is time-consuming and can be affected by inspector fatigue and variation in judgement.

This project proposes an AI-powered visual inspection system to support the inspection process. A camera observes each part as it arrives at the inspection station, a deep-learning model analyses the image and identifies the target defect classes, the inspection result is logged automatically, and a dashboard presents inspection trends. Where sufficient and reliable process data are available, the system will also support the investigation of recurring defect patterns and their possible upstream causes.

2. Problem Statement and Objective

The objective is to design and develop an AI-powered visual inspection system that detects and classifies selected surface and geometric defects on sheet-metal brackets, records the inspection results, and supports quality analysis. The system will be evaluated for detection performance, consistency and inspection speed under the defined experimental conditions, with the aim of reducing dependence on manual inspection.

2.1 Specific Objectives

- **Consistency —** Evaluate the consistency of automated defect detection under controlled inspection conditions.
- **Multi-defect coverage —** Detect and classify the selected target defects: dents or deformation, scratches, burrs, missing holes, hole-position deviations, and selected geometric deviations such as hole-diameter or bend-angle deviations, where these can be reliably measured with the calibrated inspection setup. Rust and stains will be included only if suitable labelled data are available.
- **Real-time performance —** Measure inference latency and throughput, and determine whether the proposed system can meet the defined inspection-station speed requirement.
- **Traceability —** Store inspection results together with the associated image and relevant metadata, so that rejected parts can be traced and reviewed later.
- **Root-cause insight —** Support root-cause analysis by correlating recurring defect patterns with available process information, such as die, shift or production-batch information. Any suspected relationship will be treated as an analysis result, and not as an established causal relationship, unless it is supported by evidence.
- **Dataset strategy —** Combine freely available public datasets with images collected in-house, so that pretraining on public data can compensate for the limited size of the in-house dataset.

3. Why We Chose the Sheet Metal Bracket

Choosing the right part on which to build and test the system was as important as choosing the right model. We shortlisted a few common sheet-metal components and selected a mounting or support bracket for the following practical reasons:

- **Slight but meaningful shape variation —** A bracket is neither perfectly flat nor perfectly symmetric. It has bends, flanges and a mix of hole types. This gives the model real geometric variation to learn from, unlike a flat plate on which almost any model would score well.
- **Number and pattern of holes —** Brackets typically carry multiple mounting holes of different sizes and spacing. This allows us to test the detection of both surface defects and precise, position-sensitive defects (missing hole, off-position hole and wrong diameter) on the same part.
- **Industrial relevance —** Sheet-metal brackets are widely used in electrical enclosures, automotive assemblies, HVAC systems and general fixturing. This makes the bracket a relevant component for studying automated visual inspection.
- **Manageable scope for a single-camera prototype —** A bracket can be photographed with a fixed camera and controlled lighting, which makes it suitable for a final-year prototype. The single-camera setup also limits measurements that require depth or multiple views.
- **Measurable geometric features —** Bracket dimensions such as hole diameter, hole pitch, flange length and bend angle can be defined against a reference specification. The project will measure only those geometric features that can be reliably evaluated with the calibrated inspection setup and the available reference information.
- **Availability of public data —** Bracket images are included in at least one free public metal-parts dataset (MPDD), which allows the model to be pretrained and benchmarked before the in-house data are collected.

4. System Architecture

The proposed system follows an acquisition-to-decision pipeline that represents the intended inspection workflow:

- **1. Image Acquisition —** A GigE or USB3 industrial camera, mounted at a fixed inspection position with an appropriate lens and controlled bar or strobe lighting, will capture consistent images of each part. Camera calibration will be used where geometric measurement is required, and the optical setup will be selected to minimise the relevant distortion.
- **2. Pre-processing —** Cropping, resizing, glare and shadow correction and, where needed, background subtraction will normalise the image before it reaches the model.
- **3. Defect Detection and Classification —** A trained deep-learning model (object detection or segmentation) scans the image and outputs the defect class, its location and a confidence score.

- **4. Decision and PLC Signalling —** Based on the validated inspection result, an accept or reject signal can be integrated with a PLC or another industrial interface as a deployment extension. For the prototype, the system may first demonstrate the decision signal without physical part diversion.
- **5. Logging —** Every inspection result, including the image, defect class, confidence, timestamp and (if available) the batch, die or shift ID, is written to a database.
- **6. Monitoring and Root-Cause Dashboard —** A dashboard presents pass and fail rates, defect-type trends and the available inspection metadata. Where sufficient process data are available, recurring defect patterns may be analysed against variables such as die, shift, batch or other recorded process information to support root-cause investigation.

System Architecture Diagram

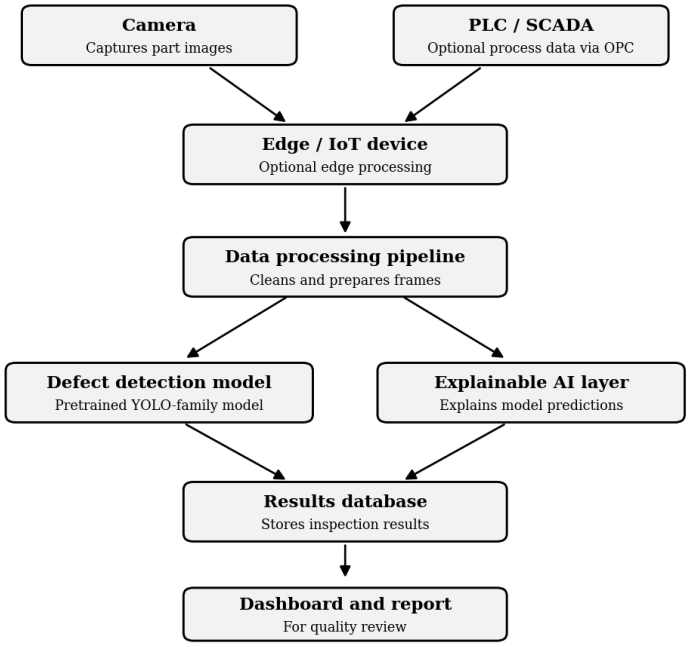


Figure 1: Proposed architecture showing camera image capture, optional process data, pre-processing, pretrained YOLO-family detection, optional explainability, result storage and dashboard reporting.

5. Methodology

5.1 Dataset

The dataset strategy combines two sources: images collected in-house under the intended inspection conditions, and free public datasets that support pretraining and benchmarking. The two sources are described below.

5.1.1 In-House Data Collection

Images of the bracket will be collected with the fixed camera and controlled lighting setup planned for inference. They will cover both acceptable parts and parts containing representative induced or naturally occurring defects. This helps the dataset reflect the intended inspection conditions, rather than only clean, staged images. Images will be labelled with bounding boxes

and, where segmentation is required, with segmentation masks. The dataset will be divided into training, validation and test sets. Augmentation, such as rotation, brightness and contrast variation and limited blur, will be evaluated to improve robustness to expected changes in lighting and part placement.

5.1.2 Public Datasets Reviewed

No single free public dataset covers all six target defects on real sheet-metal brackets. Therefore, five free public sources were reviewed and shortlisted. They will be used for pretraining, benchmarking and supporting individual defect classes, while the in-house bracket dataset remains the primary source for final training and evaluation. Table 1 summarises these sources.

Table 1: Free public datasets reviewed for this project

Dataset	Provider and Reference	Content	Access
MPDD	Jezek et al., Brno University of Technology (2021)	About 1,346 images of six metal-part categories, including three bracket types (black, brown and white), with pixel-precise defect masks. The training subset contains only defect-free images. The evaluation subset contains both normal and defective images.	GitHub repository (download link on the repository page) Licence: CC BY-NC-SA 4.0.
OSPD	Antiles and Talathi (2024)	Open Stamped Parts Dataset of stamped metal sheets for auto manufacturing. It has 1,680 labelled and 7,980 unlabelled real images from seven cameras, and a synthetic set with box and mask annotations around all holes.	arXiv:2403.10369 (download link given in the paper)
NEU-DET	Song and Yan (2013), Northeastern University	1,800 grayscale images (200 × 200 pixels) of hot-rolled steel strip surfaces in six classes: crazing, inclusion, patches, pitted surface, rolled-in scale and scratches, with 300 images per class.	Official NEU database page Roboflow copy
GC10-DET	Lv et al. (2020)	3,570 grayscale images of real industrial steel plates in ten classes: punching, weld line, crescent gap, water spot, oil spot, silk spot, inclusion, rolled pit, crease and waist folding. Some hosted versions contain 2,300 images.	Kaggle Authors' GitHub repository

Dataset	Provider and Reference	Content	Access
Roboflow Universe sets	Community contributors	Small community-labelled metal defect sets with bounding boxes that export directly to YOLO format, including a set with Scratch, Dent and other classes.	Surface defects (Scratch, Dent) Metal defect detection Surface defect detection

5.1.3 Mapping of Public Datasets to Target Defects

Table 2 shows how far the public datasets support each target defect. Burrs have no public sheet-metal data, and hole diameter and bend angle are better handled by calibrated measurement than by a learned model.

Table 2: Public data support for each target defect

Target Defect	Supporting Public Data	Comment
Dent / deformation	Roboflow (Scratch, Dent set); MPDD	Limited support. In-house dent images are required.
Scratches	NEU-DET; MPDD; Roboflow sets	Good pretraining support, although the backgrounds differ from brackets.
Burrs on edges	None found	Burrs will be induced and collected in-house.
Missing hole	OSPD; GC10-DET (punching class)	OSPD is the closest match for hole-related detection.
Hole mis-position / wrong diameter	OSPD	The final tolerance check will use calibrated OpenCV measurement.
Bend-angle deviation	MPDD (bent-part anomalies); GC10-DET (crease and waist folding)	Partial support only. The angle will be measured from contours against the CAD tolerance.

5.1.4 Data Usage Strategy

- **Pretraining —** NEU-DET and GC10-DET will be used to pretrain the detector on general metal texture and scratch features.
- **Bracket-specific baseline —** The three MPDD bracket categories will be used to build an anomaly-detection baseline (for example, PatchCore through the Anomalib library). The MPDD masks will also be converted to bounding boxes for YOLO training.
- **Hole defects —** OSPD will be used to pretrain and test the hole-detection component.
- **Fine-tuning —** The pretrained model will be fine-tuned on the in-house bracket images, which include induced burrs and hole defects.

- **Evaluation —** The final test set will contain only in-house images, so that reported results reflect the real inspection conditions. Public images will not be mixed into the test set.
- **Label harmonisation —** Class names from all sources will be mapped to one project taxonomy, all annotations will be converted to YOLO format, and duplicates will be removed. The training, validation and test splits will be checked to prevent data leakage.

5.1.5 Licensing and Limitations of Public Data

MPDD is released under the Creative Commons Attribution-NonCommercial-ShareAlike 4.0 licence, which allows non-commercial academic use with attribution. The licence terms of the other datasets will be checked individually before any results are published, and every dataset used will be cited. The public datasets were collected with different materials, cameras and lighting, so a domain gap is expected between them and the in-house setup. For this reason, the public datasets support pretraining and benchmarking only, and they do not replace in-house data collection.

5.2 Model Development

A YOLO-family detector is proposed as the primary object-detection model because it provides a practical balance between detection performance and inference speed. If precise pixel-level boundaries are required for selected defects, a segmentation approach such as U-Net may be evaluated separately. The detection and segmentation approaches will be compared using task-appropriate evaluation metrics. The selected model will be trained and validated on the labelled dataset and evaluated using precision, recall, AP and mAP, and per-class detection performance, so that weak performance on individual defect classes is not hidden by an overall average.

5.3 Optimisation for Deployment

After model training, deployment optimisation may be evaluated using ONNX Runtime, TensorRT or OpenVINO, with FP16 or INT8 quantisation where supported. The optimised model will be tested on the selected industrial PC or edge device to measure inference latency and throughput.

5.4 Explainability

Grad-CAM and SHAP-based visualisations may be explored to indicate which image regions contributed to model decisions. If implemented, these visualisations can support model interpretation, quality review and documentation. They will not, by themselves, prove that the model's decision is correct.

Inspection Workflow Diagram

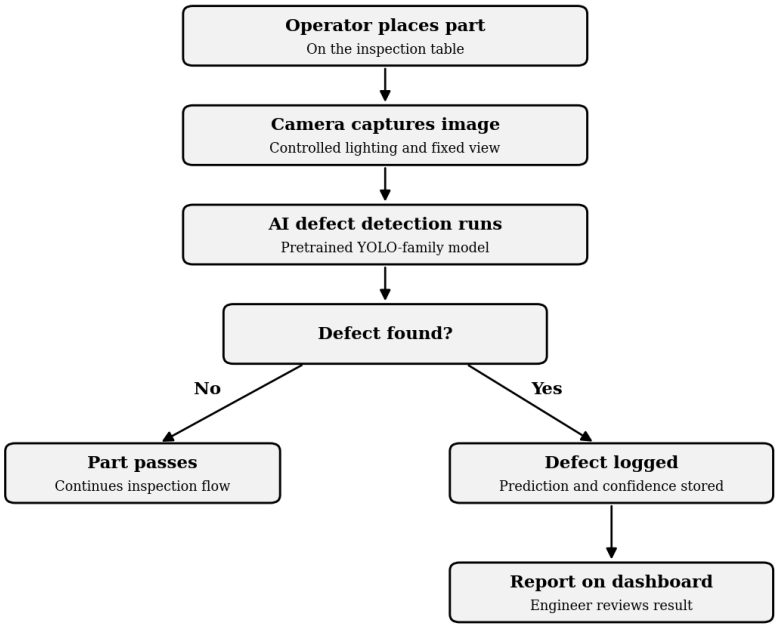


Figure 2: Proposed workflow from placing the bracket on the inspection table through image capture, AI-based defect detection, inspection decision, result logging and dashboard review.

6. Technology Stack

Table 3: Proposed tools and technologies

Category	Proposed Tools / Technologies
Languages	Python, SQL
CV / Deep Learning	PyTorch, OpenCV, YOLO-family model (YOLOv8 candidate, using the Ultralytics library)
Inference Optimisation	ONNX Runtime, TensorRT, OpenVINO
Vision Hardware	GigE or USB3 industrial camera, C-mount lens, bar or strobe lighting
Edge / Compute	NVIDIA Jetson, industrial PC
Integration	PLC (Siemens or Allen-Bradley), OPC-UA, Modbus TCP, digital I/O relay
Data and Reporting	PostgreSQL, Grafana, REST APIs
MLOps	CVAT (annotation), MLflow, Git
Dataset Tools	Roboflow (format conversion), Anomalib (anomaly-detection baseline on MPDD)

7. Defects Targeted

Table 4: Target defects and detection methods

Defect Type	Category	How It Is Detected
Dent / deformation	Surface / Forming	Object detection of localised shape or shadow anomalies
Scratches	Surface	Fine-edge texture analysis
Burrs on edges	Edge	Detection of edge-contour irregularities
Missing hole	Dimensional	Expected-hole-position check against a reference template
Hole mis-position / wrong diameter	Dimensional	Geometric measurement against a tolerance band
Bend-angle deviation	Dimensional	Contour and angle measurement against the CAD tolerance

8. Expected Outcomes and Evaluation

The project is expected to demonstrate an end-to-end prototype for automated bracket inspection under the defined experimental conditions. The evaluation will focus on detection performance, processing speed, traceability and the feasibility of integrating the inspection result with a decision interface. The intended outcomes are as follows:

- **Model performance evaluation —** Precision, recall, AP and mAP, and per-class performance will be reported for the trained defect-detection model, so that weak defect classes remain visible instead of being hidden by an overall average.
- **Public dataset benchmarking —** Results on the public datasets (MPDD, NEU-DET and GC10-DET) will be reported separately from the in-house results, so that the effect of pretraining can be measured.
- **Throughput evaluation —** Frames per second and, where applicable, inference latency will be measured before and after deployment optimisation. The measured values will be compared with the defined inspection-station speed requirement.
- **End-to-end prototype —** The project will integrate image acquisition, pre-processing, inference, result display and logging and, if implemented, a PLC or industrial-interface decision signal.
- **Monitoring dashboard —** A dashboard will present inspection results, pass and fail rates and defect-type trends to support quality monitoring.
- **Explainability support —** If Grad-CAM or SHAP is implemented, the visualisations will be associated with selected inspection results to support interpretation and documentation. They will not be treated as proof of causal reasoning.

- **Root-cause analysis support —** Where sufficient historical process data are available, defect trends will be analysed against relevant process variables. Any relationships identified will be reported as associations, unless additional evidence supports causality.
- **Business and operational evaluation —** The project will compare the measured inspection performance with the defined manual-inspection requirements and will discuss the potential benefits and limitations of AI-assisted inspection. Claims of improvement will be based on measured results.

9. Advantages of This Approach

- **Consistency —** The automated system is intended to provide repeatable inspection results under controlled conditions. The variation will be evaluated experimentally.
- **Potential for 100% inspection coverage —** Continuous automated inspection can evaluate every part presented to the inspection station, subject to the actual capture rate, system availability and line conditions.
- **Potential throughput improvement —** The system may support faster inspection, but any improvement will be claimed only after the system latency and inspection rate are measured against the defined requirement.
- **Traceability —** Each stored inspection result can include the image, the model output and confidence, the timestamp and the available production metadata. Explainability overlays will be included only if the explainability module is implemented.
- **Reduced data-collection effort —** Pretraining on free public datasets reduces the number of in-house images that must be collected and labelled.
- **Process improvement, not just sorting —** Patterns in the defect data feed back into the process, and not only into a reject bin.

10. Limitations and Future Scope

The performance of the model will depend strongly on dataset coverage and label quality. Rare or newly introduced defect types may require additional labelled data and periodic retraining. The public datasets do not cover burrs on sheet-metal brackets, and they differ from the in-house setup in material, camera and lighting, so in-house data collection remains essential. Lighting consistency is another important limitation. The prototype assumes controlled lighting, and changes in camera position or ambient conditions may require recalibration or additional data collection.

As future work, the system can be extended to multiple part types beyond the bracket, using a shared backbone with part-specific heads. It can also be extended to multi-camera, 3D or depth-based inspection to catch dimensional defects that a single 2D image can miss, and to a closed-loop retraining pipeline in which the model improves itself from newly logged production images.

11. Conclusion

This project focuses on developing a practical AI-based inspection system for sheet-metal brackets. The main idea is to use a camera to capture the bracket, process the image and identify selected defects such as dents, scratches, burrs, missing holes and geometric deviations. Instead of depending completely on manual inspection, the proposed system is intended to make the inspection process more consistent and easier to monitor.

The project also brings the different parts of an inspection system together. The captured images will be processed and given to a pretrained YOLO-family model for defect detection. Free public datasets (MPDD, OSPD, NEU-DET, GC10-DET and Roboflow Universe sets) will support pretraining and benchmarking, and the in-house bracket dataset will be used for final training and evaluation. The inspection results can then be stored and displayed through a dashboard for quality review. Where suitable process information is available, the recorded defect patterns can also be studied to support the investigation of possible causes. The system will be evaluated using measures such as precision, recall, AP and mAP, per-class performance and inference speed.

Overall, the aim of the project is not only to detect whether a bracket has a defect, but also to build a complete and understandable inspection workflow from image capture to reporting. The proposed prototype can provide a starting point for future industrial use. Further work with larger datasets, more varied inspection conditions, proper calibration and additional process data would be needed before the system is used in a real production environment.

12. References and Dataset Sources

[1] S. Jezek, M. Jonak, R. Burget, P. Dvorak and M. Skotak, "Deep learning-based defect detection of metal parts: evaluating current methods in complex conditions," in Proc. 13th International Congress on Ultra Modern Telecommunications and Control Systems and Workshops (ICUMT), 2021, pp. 66–71, doi: 10.1109/ICUMT54235.2021.9631567. Dataset (MPDD): https://github.com/stepanje/MPDD

[2] S. Antiles and S. Talathi, "Open Stamped Parts Dataset," arXiv:2403.10369, 2024. https://arxiv.org/abs/2403.10369

[3] K. Song and Y. Yan, "A noise robust method based on completed local binary patterns for hot-rolled steel strip surface defects," Applied Surface Science, 2013. Dataset (NEU-DET): https://faculty.neu.edu.cn/songkc/en/zdylm/263265/list/index.htm

[4] X. Lv, F. Duan, J. Jiang, X. Fu and L. Gan, "Deep metallic surface defect detection: the new benchmark and detection network," Sensors, 2020. Dataset (GC10-DET): https://github.com/lvxiaoming2019/GC10-DET-Metallic-Surface-Defect-Datasets and https://www.kaggle.com/datasets/lirick/gc10-det

[5] Roboflow Universe, Surface defects dataset (scratch): https://universe.roboflow.com/scratch-8kuvz/surface-defects-2sby7-j8p3p

[6] Roboflow Universe, Metal defect detection dataset (capjamesg): https://universe.roboflow.com/capjamesg/metal-defect-detection-d3wl6

[7] Roboflow Universe, Surface defect detection dataset (METAL SURFACE DEFECTS): https://universe.roboflow.com/metal-surface-defects/surface-defect-detection-boivj